principle," and that speaking about undetermined quantities in quantum physics was "neither true nor false, but *meaningless*"—claims that sound awfully similar to the Copenhagen interpretation.

None of this is to say that logical positivism was the philosophical motivation behind the Copenhagen interpretation. Bohr, in particular, may not have been much of a positivist. While it's hard to say what Bohr's position actually was—there's a small mountain of papers attempting to decipher his views on any given subject, and few of them agree—Bohr did seem to flirt with certain ideas that the positivists found distasteful, such as vitalism. (In fact, Bohr spoke favorably about vitalism with complementarity-based reasoning at the 1936 conference at his own house, while Schlick's aforementioned paper at the same conference argued against vitalism.) And Neurath thought that Bohr's "printed remarks are full of crass metaphysics," and that he "express[es] himself somewhat unclearly." But Bohr also seemed sympathetic to the positivists, and at times came close to declaring himself one of them. When Frank asked whether Bohr's reply to the EPR paper was based on positivist reasoning, Bohr told him that "you have caught the sense of my efforts very well."

Regardless of Bohr's true philosophical convictions, the Copenhagen interpretation was certainly defended with arguments and slogans derived from logical positivism. The verification theory of meaning—especially the idea that unverifiable statements are meaningless—was introduced to physics students as a fundamentally new insight about how the world worked, an inextricable part of the success of quantum physics. According to a very popular midcentury quantum physics textbook, the old physics before the quantum revolution assumed that particles like photons always had a definite position at every moment of time, but "quantum mechanics… asserts instead that the position of a photon has meaning only when the experiment includes a position determination." Heisenberg himself often used operationalist language when talking about the quantum world. "There is no way of observing the orbit of the electron around the nucleus" of an atom, he declared, and "therefore, there is no orbit in the ordinary sense." According to him, presuming the electron has an orbit or any kind of path at all between observations "would be a misuse of language which… cannot be justified."

Yet, in truth, the physics community hadn't really embraced positivism